TD2: Depth in Neural networks

<u>Objectives: build a deep neural network with Keras</u>

<u>Exercise 1 : Build a deep neural network with Keras</u>

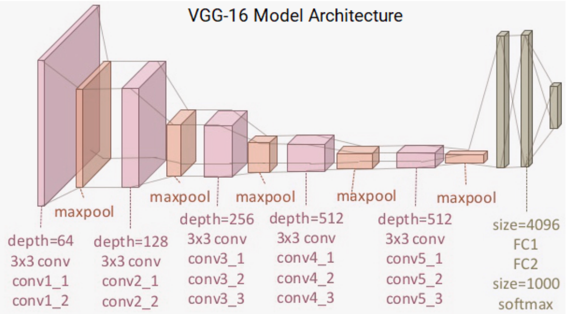
<u>Objectif :</u> Use Keras to build a convolution neural net and train it on a off-the-shelf classification task (Cifar10).

1. Starting from the code you build on the exercise 3 of the first tutorial, build and format the Cifar10 dataset so that it can be used by a neural network. This time, make sure that the images are kept as is, with all 3 color channels (RGB). Don't forget to split the dataset in train/test sets!
2. Build a neural network with the following architecture:
    a. A 3x3 convolution layer, depth 32 with an ReLU activation function.
    b. A 3x3 convolution layer, depth 64 with an ReLU activation function.
    c. A 2x2 maxpool layer with a dropout of 0.25.
    d. A 3x3 convolution layer, depth 128 with a ReLU activation function.
    e. A 2x2 maxpool layer (no dropout).
    f. A 3x3 convolution layer, depth 128 with a ReLU activation function.
    g. A 2x2 maxpool layer with a dropout of 0.25.
    h. A flatten layer
    i. A 1024-neurons dense layer with ReLU activation function and dropout of 0.5.
    j. A 10-neurons dense layer with softmax activation function (no dropout).
3. Compile the network using:
    a. Adam optimizer
    b. Learning rate of 0.0001
    c. Decay of 10^{-6}
4. Train the network for 100 epochs with a batch size of 32. It can take time, specially with CPU only
5. Evaluate the network on the test dataset. What do you think about the results, especially in comparison the multilayer perceptron?

<u>Exercise 2 : Use/implement an existing architecture.</u>

<u>Objective:</u> Implement a known network architecture and train on an off-the-shelf dataset.

1. Look for the VGG16 architecture on the internet. Implement this architecture using Keras.



2. Train and evaluate the network on Cifar10 dataset.
3. Train and evaluate the network on Cifar100. What do you think about the results?
4. Look over the internet for a efficient architecture for this problem. Implement this architecture using Keras.
5. Look at the Kaggle website (www.kaggle.com) and find a image analysis problem to solve with deep learning)